ChE 210 Homework Set #7 Due on Tuesday, 10 March 2020

(1) Ethane is chlorinated in a continuous reactor:

$$C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$$

Some of the product monochloroethane is further chlorinated in an undesired side reaction:

$$C_2H_5Cl + Cl_2 \rightarrow C_2H_4Cl_2 + HCl$$

(a) Suppose your objective is to maximize selectivity of the monochloroethane production relative to the dichloroethane production. Would you design the reactor for a high or low conversion of ethane? Explain your answer. What additional processing steps would almost certainly be necessary to make the process economical?

(b) Draw and label a flowchart, assuming that the reactor feed contains only the two reactants. Use a degree-of-freedom analysis based on atomic species balances to determine how many process variable values must be specified for you to be able to calculate the remainder.

(c) The reactor is designed to yield a 15% conversion of ethane and a selectivity of 14 moles of C_2H_5Cl for every mole of $C_2H_4Cl_2$. Assume all of the chlorine gas is reacted. Calculate the feed ratio (moles Cl_2 /moles C_2H_6) and the fractional yield of monochloroethane.

(d) Suppose the reactor is built and started up and the conversion is only 14%. Chromatographic analysis shows that there is no Cl_2 in the product, but another species of molecular weight higher that that of dichloroethane is present. Offer a likely explanation of the results.

(2) Murphy 4.40

(3) Murphy 4.49

(4) Murphy 4.50